Sophie Yifei **Wang**

CURRICULUM VITAE

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AI Agents · Reinforcement Learning · Memory & Self-Evolution

Education

Beijing Institute of Technology *Beijing, China*

B.ENG. CANDIDATE IN COMPUTER SCIENCE AND TECHNOLOGY *Sep. 2024 – Present*

- Xu Teli Honors Program; class representative.
- CET-4: 572; Second Prize in the National English Competition for College Students.

Research Interests

Adaptive Agents: reinforcement learning with long-term memory and long-horizon decision-making. **Self-Evolving Models:** recursive learning, feedback-driven self-improvement, and capability evolution. **Multimodal Intelligence:** multimodal foundation models, agentic systems, model interpretability, and brain–model semantic alignment.

Research Experience

Threader · Long-term Memory for Conversational Agents *Research collaboration with Tsinghua University*

CO-AUTHOR · NEURIPS 2026 UNDER REVIEW *Dec. 2025 – May 2026*

- Co-developed Threader, a lossless long-term memory framework addressing information loss, latency, and cost in compression-based memory systems.
- Introduced a topic-segmentation, multi-view encoding, and multi-signal retrieval pipeline that preserves complete conversations while improving construction and retrieval efficiency and reducing token cost.
- Outperformed representative systems including MemGPT and Mem0 on long-conversation memory benchmarks, improving long-range consistency and response efficiency.

SAIL · Brain–Model Semantic Alignment *Beijing Institute of Technology*

CO-AUTHOR · NEURIPS 2026 UNDER REVIEW *Sep. 2025 – May 2026*

- Co-developed a sparse autoencoder (SAE) framework for aligning brain activity with interpretable representations inside multimodal foundation models.
- Disentangled high-dimensional visual features into monosemantic concepts and matched them to fMRI signals from the human visual cortex, revealing hierarchical correspondences with the ventral visual pathway.
- Demonstrated that SAE features outperform raw model features in brain-alignment tasks.

Industry Experience

ByteDance · Volcano Ark *Beijing, China*

PRODUCT SOLUTIONS INTERN · AI PRODUCTS & MAAS SOLUTIONS *Apr. 2026 – Jun. 2026*

- Contributed to enterprise AI product and MaaS solution design by mapping foundation-model capabilities to business scenarios and product delivery paths.
- Supported solution research and product communication for B2B AI use cases, translating technical capabilities into actionable business proposals.

Selected Projects

Multi-Agent Shopping Assistant

GitHub

Python · Vue 3 · MCP · RAG — *Nov. 2025 – Present*

- Built an e-commerce assistant around the Model Context Protocol, implementing the LLM → Router → MCP Server → Backend API tool-calling pipeline.
- Explored a Planner–Worker multi-agent architecture and developed a Vue 3 conversational interface that renders structured tool results as visual product cards.

Collector+ (Collector-AI)

GitHub · Live Demo

FastAPI · React · LLM · Prompt Engineering — *Jan. 2026*

- Built a low-latency information extraction and gamified-reading assistant that transforms long-form articles and videos into binary prediction questions and supports source-grounded Q&A over saved content.
- Designed a linear-monolithic asynchronous pipeline with approximately 15–30 seconds of end-to-end generation latency and general-purpose parsing for WeChat articles, Bilibili videos, and dynamically rendered pages.
- Implemented structured prompt engineering and an intuition-driven interaction flow with React Context state management.

FitSnap

Live Demo

Next.js 16 · React 19 · DeepSeek-V3 · Zustand — *Dec. 2025*

- Transformed unstructured instructional videos into structured, timestamped steps with action categories and key parameters using schema-constrained JSON output.
- Built a content-agnostic generative UI with bidirectional synchronization between video playback and task-specific step cards for fitness, cooking, and other instructional content.

BiGEM Data Retrieval and Visualization Platform

Source · Project Wiki

React · D3.js · Three.js · Python — *Sep. 2024 – Oct. 2025*

- Collected, cleaned, and connected iGEM data from 2004–2024 and visualized global collaboration networks using force-directed graphs, trees, and Sankey diagrams.
- Integrated LLM-based semantic analysis to improve retrieval of biological parts and historical projects; received a 2025 iGEM Gold Medal and Best Software in the undergraduate category.

Technical Skills

Programming & ML	Python, PyTorch, TypeScript
Development	React, Git
Languages	Chinese (Native), English (fluent)

Honors & Awards

2026	**Honorable Mention**, Mathematical Contest in Modeling / Interdisciplinary Contest in Modeling	
2025	**Gold Medal & Best Software**, iGEM Competition · Undergraduate Category	
2025	**Beijing First Prize**, China Undergraduate Mathematical Contest in Modeling	*Beijing, China*
2025	**Second Prize**, National English Competition for College Students	